15-441 15-641 Computer Networking

NATs, VPN, Translaion

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www.cs.cmu.edu/~prs/15-441-F13

Outline

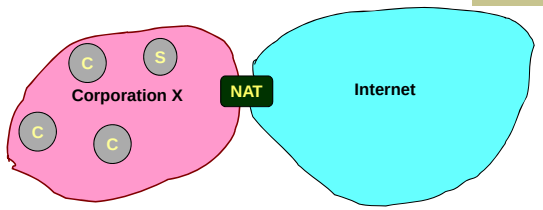
- NAT
- Tunneling and VPNs
- Name and address translation

Altering the Addressing Model

- Original IP Model: Every host has unique IP address
- Implications
  - Any host can communicate with any other host
  - Any host can act as a server
    - Just need to know host ID and port number
- System is open – complicates security
  - Any host can attack any other host
  - Possible to forge packets
    - Use invalid source address
- Places pressure on the address space
  - Every host requires "public" IP address

Reducing IP Addresses

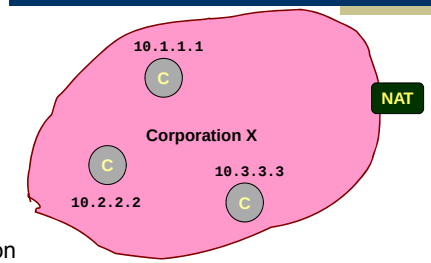
C: Client
S: Server



- Most machines within organization are used by individuals
  - For most applications, they act as clients
- Small number of machines act as servers for entire organization
  - E.g., mail server, web, ..
  - All traffic to outside passes through firewall

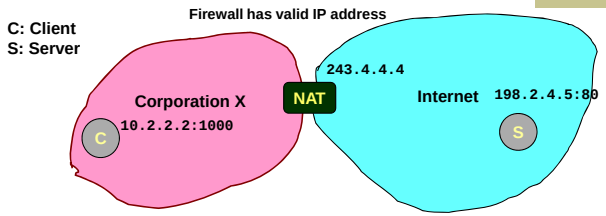
(Most) machines within organization do not need public IP addresses!

Network Address Translation (NAT)



- Within Organization
  - Assign every host an unregistered IP address
    - IP addresses 10/8 & 192.168/16 unassigned
  - Route within organization by IP protocol, can do subnetting, …
- Firewall
  - Does not let any packets from internal nodes escape
  - Outside world does not need to know about internal addresses

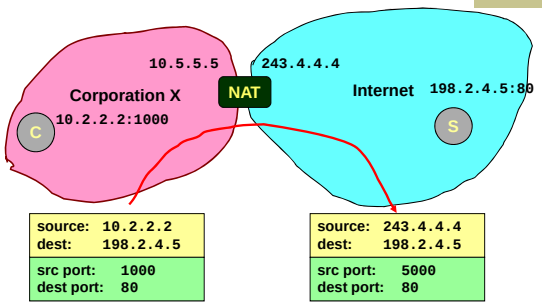
NAT: Opening Client Connection



- Client 10.2.2.2 wants to connect to server 198.2.4.5:80
  - OS assigns ephemeral port (1000)
- Connection request intercepted by firewall
  - Maps client to port of firewall (5000)
  - Creates NAT table entry

Int Addr	Int Port	NAT Port
10.2.2.2	1000	5000

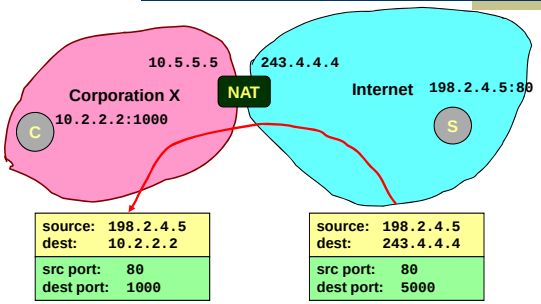
NAT: Client Request



Int Addr	Int Port	NAT Port
10.2.2.2	1000	5000

- Firewall acts as proxy for client
  - Intercepts message from client and marks itself as sender

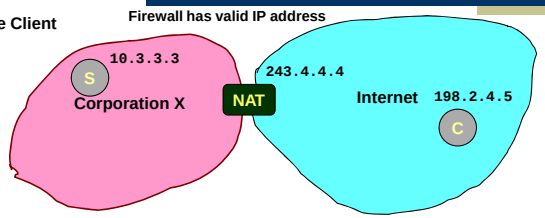
NAT: Server Response



Int Addr	Int Port	NAT Port
10.2.2.2	1000	5000

- Firewall acts as proxy for client
  - Acts as destination for server messages
  - Relabels destination to local addresses

NAT: Enabling Servers



- Use port mapping to make servers available

Int Addr	Int Port	NAT Port
10.3.3.3	80	80

  - Manually configure NAT table to include entry for well-known port
  - External users give address 243.4.4.4:80
  - Requests forwarded to server

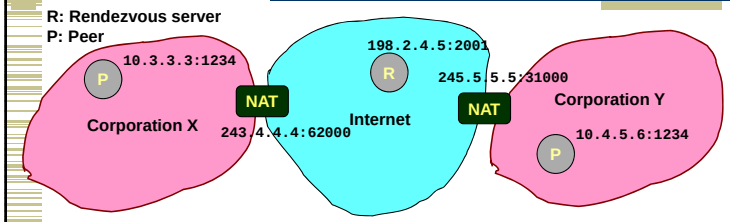
NAT Considerations

- NAT has to be consistent during a session.
  - Set up mapping at the beginning of a session and maintain it during the session
    - Recall 2nd level goal 1 of Internet: Continue despite loss of networks or gateways
    - What happens if your NAT reboots?
  - Recycle the mapping that the end of the session
    - May be hard to detect
- NAT only works for certain applications.
  - Some applications (e.g. ftp) pass IP information in payload
  - Need application level gateways to do a matching translation
  - Peer-peer apps, multi-player games have problems – who is the server?
- NAT is loved and hated
  - Breaks some applications
  - Inhibits deployment of new applications like p2p (but so do firewalls!)
  + Little NAT boxes make home networking simple.
  + Saves addresses. Makes allocation simple.

Properties of Firewalls with NAT

- Advantages
  - Hides IP addresses used in internal network
    - Easy to change ISP: only NAT box needs to have IP address
    - Fewer registered IP addresses required
  - Basic protection against remote attack
    - Does not expose internal structure to outside world
    - Can control what packets come in and out of system
    - Can reliably determine whether packet from inside or outside
- Disadvantages
  - Contrary to the “open addressing” scheme envisioned for IP addressing
  - Hard to support peer-to-peer applications
    - Why do so many machines want to serve port 1214?

Many Options Exist



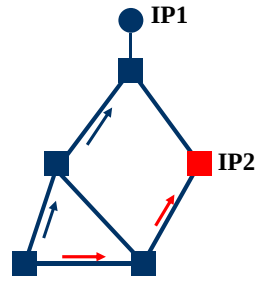
- NAT recognizes certain protocols and behaves as a application gateway
  - Used for standard protocols such as ftp
- Applications negotiate directly with NAT or firewall – need to be authorized
  - Multiple protocols dealing with different scenarios
- Punching holes in NAT: peers contact each other simultaneously using a known public (IP, port), e.g. used with rendezvous service
  - Use publicly accessible rendezvous service to exchange accessibility information
  - Assumes NATs do end-point independent mapping

Outline

- NAT
- Tunneling and VPNs
- Name and address translation

Tunneling

- Force a packet to go to a specific point in the network.
  - Path taken is different from the regular routing
- Achieved by adding an extra IP header to the packet with a new destination address.
  - Similar to putting a letter in another envelope
  - preferable to using IP source routing option
- Used increasingly to deal with special routing requirements or new features.
  - Mobile IP,..
  - Multicast, IPv6, research, ..



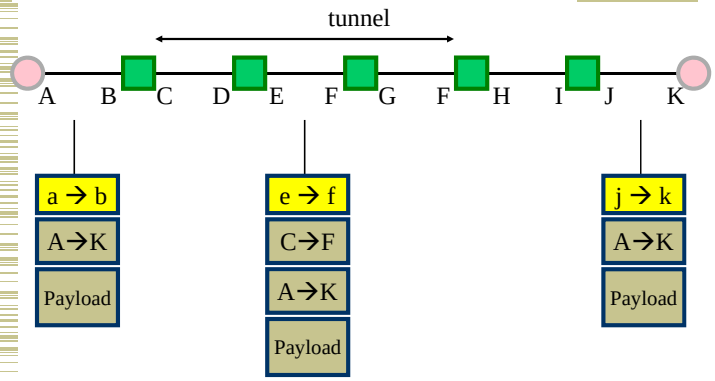
Data	IP1	IP2

IP-in-IP Tunneling

- Described in RFC 1993.
- IP source and destination address identify tunnel endpoints.
- Protocol id = 4.
  - IP
- Several fields are copies of the inner-IP header.
  - TOS, some flags, ..
- Inner header is not modified, except for decrementing TTL.

V/HL	TOS	Length
ID		Flags/Offset
TTL	4	H. Checksum
Tunnel Entry IP		
Tunnel Exit IP		
V/HL	TOS	Length
ID		Flags/Offset
TTL	Prot.	H. Checksum
Source IP address		
Destination IP address		
Payload		

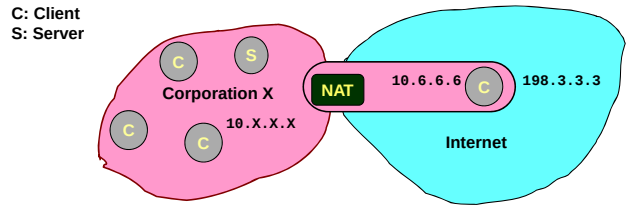
Tunneling Example



Tunneling Applications

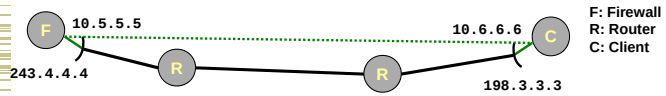
- Virtual private networks.
  - Connect subnets of a corporation using IP tunnels
  - Often combined with IP Sec (later)
- Support for new or unusual protocols.
  - Routers that support the protocols use tunnels to "bypass" routers that do not support it
  - E.g. multicast, IPv6 (!)
- Force packets to follow non-standard routes.
  - Routing is based on outer-header
  - E.g. mobile IP (later)

Extending Private Network



- Supporting Road Warrior
  - Employee working remotely with assigned IP address 198.3.3.3
  - Wants to appear to rest of corporation as if working internally
    - From address 10.6.6.6
    - Gives access to internal services (e.g., ability to send mail)
- Virtual Private Network (VPN)
  - Overlays private network on top of regular Internet

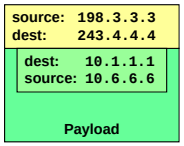
Supporting VPN by Tunneling



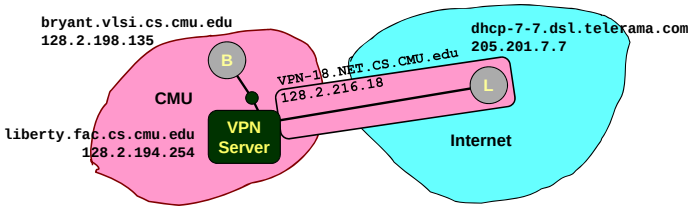
- Concept
  - Appears as if two hosts connected directly
- Usage in VPN
  - Create tunnel between client & firewall
  - Remote client appears to have direct connection to internal network

Supporting VPN by Tunneling

- Client creates packet for internal node 10.1.1.1
- Entering Tunnel
  - Add extra IP header directed to firewall (243.4.4.4)
  - Original header becomes part of payload
  - Possible to encrypt it
- Exiting Tunnel
  - Firewall receives packet
  - Strips off header
  - Sends through internal network to destination



CMU CS VPN Example



- CS has server to provide VPN services
- Operation
  - Running echo server on CMU machine 128.2.198.135
  - Run echo client on laptop connected through DSL from non-CMU ISP
- With VPN
  - server connected to 128.2.216.18 - VPN-18.NET.CS.CMU.EDU
- Without VPN
  - server connected to 205.201.7.7 - dhcp-7-7.dsl.telerama.com
- Effect
  - For CMU hosts, packets appear to originate from within CMU

Overlay Networks

- A network "on top of the network".
  - E.g., initial Internet deployment
    - Internet routers connected via phone lines
      - An overlay on the phone network
  - Tunnels between nodes on a current network
- Examples: IPv6 "6bone", multicast "Mbone".
- But not limited to IP-layer protocols…
  - Peer-to-peer networks, anonymising overlays
  - Application layer multicast
  - Improve routing, e.g. work around route failures

Important Concepts

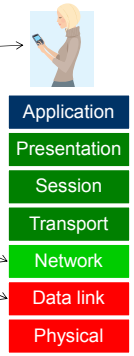
- Changes to Addressing Model
  - Have moved away from "everyone knows everybody" model of original Internet
- Firewalls + NAT hide internal networks
- VPN / tunneling build private networks on top of commodity network
- IPv6
  - Cleanup of various v4 flaws
  - Larger addresses

Outline

- NAT
- Tunneling and VPNs
- Name and address translation
  - ARP
  - DNS

Too Much of a Good Thing?

- Hosts have a
  - host name
  - IP address
  - MAC address
- There is a reason ..
  - Remember?
- But how do we translate?



IP to MAC Address Translation

- How does one find the Ethernet address of a IP host?
- ARP
  - Broadcast search for IP address
    - E.g., “who-has 128.2.184.45 tell 128.2.206.138” sent to Ethernet broadcast (all FF address)
  - Destination responds (only to requester using unicast) with appropriate 48-bit Ethernet address
    - E.g, “reply 128.2.184.45 is-at 0:d0:bc:f2:18:58” sent to 0:c0:4f:d:ed:c6

Caching ARP Entries

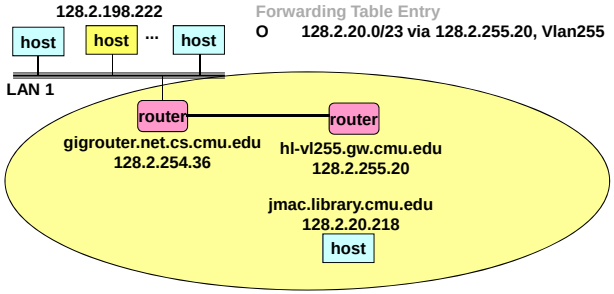
- Efficiency Concern
  - Would be very inefficient to use ARP request/reply every time need to send IP message to machine
- Each Host Maintains Cache of ARP Entries
  - Add entry to cache whenever get ARP response
  - Set timeout of ~20 minutes

ARP Cache Example

- Show using command “arp -a”

```
Interface: 128.2.222.198 on Interface 0x1000003
  Internet Address      Physical Address      Type
  128.2.20.218          00-b0-8e-83-df-50     dynamic
  128.2.102.129         00-b0-8e-83-df-50     dynamic
  128.2.194.66          00-02-b3-8a-35-bf     dynamic
  128.2.198.34          00-06-5b-f3-5f-42     dynamic
  128.2.203.3           00-90-27-3c-41-11     dynamic
  128.2.203.61          08-00-20-a6-ba-2b     dynamic
  128.2.205.192         00-60-08-1e-9b-fd     dynamic
  128.2.206.125         00-d0-b7-c5-b3-f3     dynamic
  128.2.206.139         00-a0-c9-98-2c-46     dynamic
  128.2.222.180         08-00-20-a6-ba-c3     dynamic
  128.2.242.182         08-00-20-a7-19-73     dynamic
  128.2.254.36          00-b0-8e-83-df-50     dynamic
```

CMU's Internal Network Structure

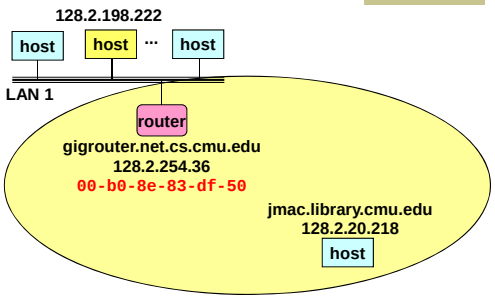


Forwarding Table Entry

O 128.2.20.0/23 via 128.2.255.20, Vlan255

- CMU Uses Routing Internally
  - Maintains forwarding tables using OSPF
  - Most CMU hosts cannot be reached at link layer

Proxy ARP



- Provides Link-Layer Connectivity Using IP Routing
  - Local router (gigrouter) sees ARP request
  - Uses IP addressing to locate host
  - Becomes "Proxy" for remote host
    - Using its own MAC address
  - Requestor thinks that it is communicating directly with remote host

- DNS

Naming

- How do we efficiently locate resources?
  - DNS: name → IP address
- Challenge
  - How do we scale this to the wide area?

Obvious Solutions (1)

Why not centralize DNS?

- Single point of failure
- Traffic volume
- Distant centralized database
- Single point of update

- Doesn't *scale!*

Obvious Solutions (2)

Why not use /etc/hosts?

- Original Name to Address Mapping
  - Flat namespace
  - /etc/hosts
  - SRI kept main copy
  - Downloaded regularly
- Count of hosts was increasing: machine per domain → machine per user
  - Many more downloads
  - Many more updates

Domain Name System Goals

- Basically a wide-area distributed database
- Scalability
- Decentralized maintenance
- Robustness
- Global scope
  - Names mean the same thing everywhere
- Don't need
  - Atomicity
  - Strong consistency

Programmer's View of DNS

- Conceptually, programmers can view the DNS database as a collection of millions of *host entry structures*:

```
/* DNS host entry structure */
struct addrinfo {
        int     ai_family;          /* host address type (AF_INET) */
        size_t  ai_addrlen;         /* length of an address, in bytes */
        struct sockaddr *ai_addr;   /* address! */
        char    *ai_canonname;      /* official domain name of host */
        struct addrinfo *ai_next;   /* other entries for host */
};
```

- Functions for retrieving host entries from DNS:
  - `getaddrinfo:` query key is a DNS host name.
  - `getnameinfo:` query key is an IP address.

DNS Message Format

12 bytes	Identification	Flags
	No. of Questions	No. of Answer RRs
	No. of Authority RRs	No. of Additional RRs
Name, type fields for a query	Questions (variable number of answers)	
RRs in response to query	Answers (variable number of resource records)	
Records for authoritative servers	Authority (variable number of resource records)	
Additional "helpful info that may be used	Additional Info (variable number of resource records)	

DNS Header Fields

- Identification
  - Used to match up request/response
- Flags
  - 1-bit to mark query or response
  - 1-bit to mark authoritative or not
  - 1-bit to request recursive resolution
  - 1-bit to indicate support for recursive resolution

DNS Records

RR format: **(class, name, value, type, ttl)**

- DB contains tuples called resource records (RRs)
  - Classes = Internet (IN), Chaosnet (CH), etc.
  - Each class defines value associated with type

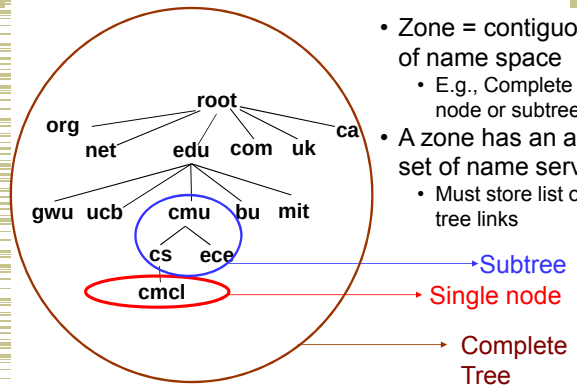
FOR IN class:

- Type=A
  - **name** is hostname
  - **value** is IP address
- Type=NS
  - **name** is domain (e.g. foo.com)
  - **value** is name of authoritative name server for this domain
- Type=CNAME
  - **name** is an alias name for some "canonical" (the real) name
  - **value** is canonical name
- Type=MX
  - **value** is hostname of mailserver associated with **name**

Properties of DNS Host Entries

- Different kinds of mappings are possible:
  - Simple case: 1-1 mapping between domain name and IP addr:
    - `kittyhawk.cmcl.cs.cmu.edu` maps to `128.2.194.242`
  - Multiple domain names maps to the same IP address:
    - `eecs.mit.edu` and `cs.mit.edu` both map to `18.62.1.6`
  - Single domain name maps to multiple IP addresses:
    - `aol.com` and `www.aol.com` map to multiple IP addrs.
  - Some valid domain names don't map to any IP address:
    - for example: `cmcl.cs.cmu.edu`

DNS Design: Zone Definitions



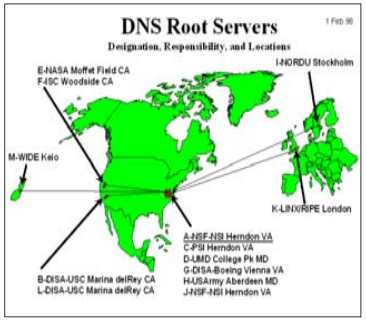
- Zone = contiguous section of name space
  - E.g., Complete tree, single node or subtree
- A zone has an associated set of name servers
  - Must store list of names and tree links

DNS Design: Management

- Zones are created by convincing owner node to create/delegate a subzone
  - Records within zone stored multiple redundant name servers
  - Primary/master name server updated manually
  - Secondary/redundant servers updated by zone transfer of name space
    - Zone transfer is a bulk transfer of the "configuration" of a DNS server – uses TCP to ensure reliability
- Example:
  - CS.CMU.EDU created by CMU.EDU administrators
  - Who creates CMU.EDU or .EDU?

DNS: Root Name Servers

- Responsible for "root" zone
- Approx. 13 root name servers worldwide
  - Currently {a-m}.root-servers.net
  - Very well protected
- Local name servers contact root servers when they cannot resolve a name
  - Configured with well-known root servers
  - Newer picture → www.root-servers.org



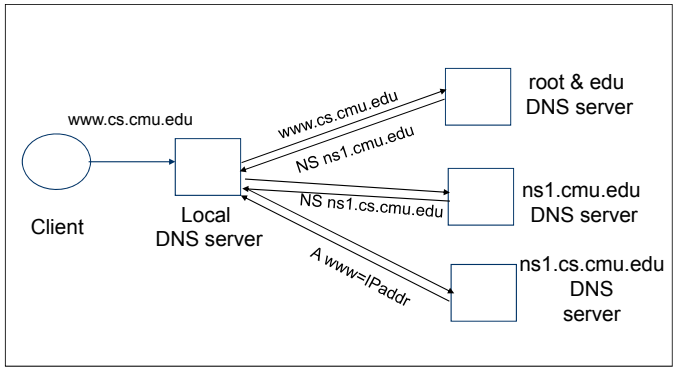
Root Zone

- Generic Top Level Domains (gTLD) = .com, .net, .org, etc…
- Country Code Top Level Domain (ccTLD) = .us, .ca, .fi, .uk, etc…
- Root server ({a-m}.root-servers.net) also used to cover gTLD domains
  - Load on root servers was growing quickly!
  - Moving .com, .net, .org off root servers was clearly necessary to reduce load → done Aug 2000

Servers/Resolvers

- Each host has a resolver
  - Typically a library that applications can link to
  - Local name servers hand-configured (e.g. /etc/resolv.conf)
- Name servers
  - Either responsible for some zone or…
  - Local servers
    - Do lookup of distant host names for local hosts
    - Typically answer queries about local zone

Typical Resolution



Typical Resolution

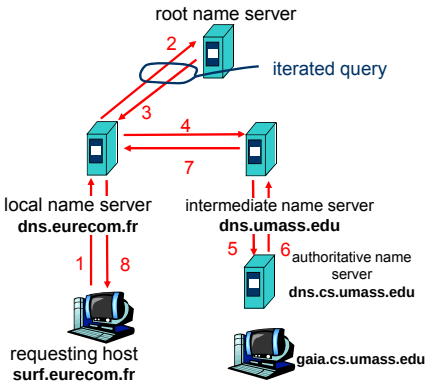
- Steps for resolving www.cmu.edu
  - Application calls gethostbyname() (RESOLVER)
  - Resolver contacts local name server (S_1)
  - S_1 queries root server (S_2) for (www.cmu.edu)
  - S_2 returns NS record for cmu.edu (S_3)
  - What about A record for S_3?
    - This is what the additional information section is for (PREFETCHING)
  - S_1 queries S_3 for www.cmu.edu
  - S_3 returns A record for www.cmu.edu

Lookup Methods

Recursive query:
- Server goes out and searches for more info (recursive)
- Only returns final answer or "not found"

Iterative query:
- Server responds with as much as it knows (iterative)
- "I don't know this name, but ask this server"

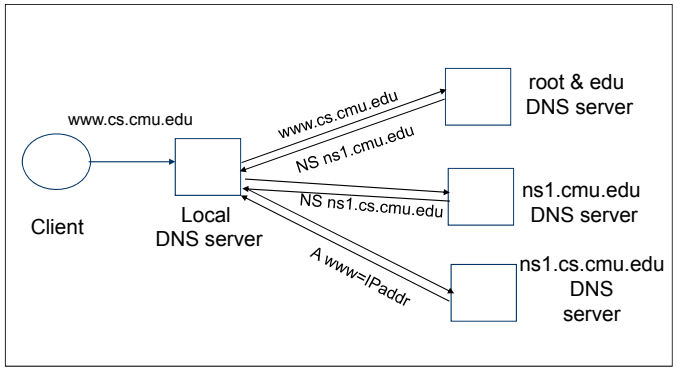
Workload impact on choice?
- Local server typically does recursive
- Root/distant server does iterative



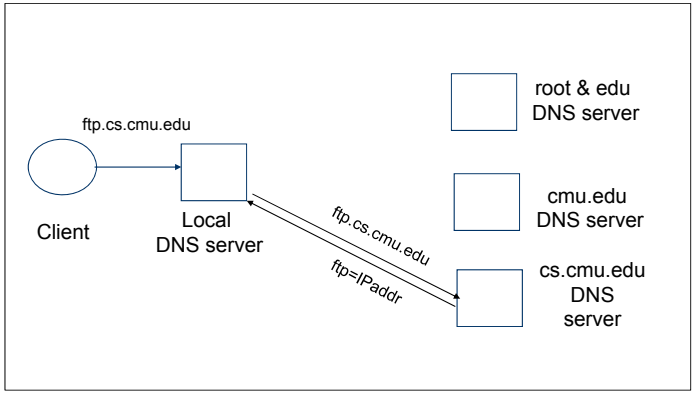
Workload and Caching

- Are all servers/names likely to be equally popular?
  - Why might this be a problem? How can we solve this problem?
- DNS responses are cached
  - Quick response for repeated translations
  - Other queries may reuse some parts of lookup
- DNS negative queries are cached
  - Don't have to repeat past mistakes, e.g., misspellings
- Cached data periodically times out
  - Lifetime (TTL) of data controlled by owner of data
  - TTL passed with every record
- Responses can include additional information
  - Often used for prefetching, e.g., CNAME/MX/NS records

Typical Resolution



Subsequent Lookup Example



Reliability

- DNS servers are replicated
  - Name service available if ≥ one replica is up
  - Queries can be load balanced between replicas
  - Queries return multiple A records
- UDP used for queries
  - Need reliability → must implement this on top of UDP!
  - Why not just use TCP?
- Try alternate servers on timeout
  - Exponential backoff when retrying same server
- Same identifier for all queries
  - Don't care which server responds

Mail Addresses

- MX records point to mail exchanger for a name
  - E.g. mail.acm.org is MX for acm.org
- Addition of MX record type proved to be a challenge
  - How to get mail programs to lookup MX record for mail delivery?
  - Needed critical mass of such mailers

Tracing Hierarchy (1)

- Dig Program
  - Allows querying of DNS system
  - Use flags to find name server (NS)
  - Disable recursion so that operates one step at a time

```
unix> dig +norecurse @a.root-servers.net NS kittyhawk.cmcl.cs.cmu.edu

;; AUTHORITY SECTION:
edu.        172800  IN    NS    L3.NSTLD.COM.
edu.        172800  IN    NS    D3.NSTLD.COM.
edu.        172800  IN    NS    A3.NSTLD.COM.
edu.        172800  IN    NS    E3.NSTLD.COM.
edu.        172800  IN    NS    C3.NSTLD.COM.
edu.        172800  IN    NS    F3.NSTLD.COM.
edu.        172800  IN    NS    G3.NSTLD.COM.
edu.        172800  IN    NS    B3.NSTLD.COM.
edu.        172800  IN    NS    M3.NSTLD.COM.
```


DNS (Summary)

- Motivations → large distributed database
  - Scalability
  - Independent update
  - Robustness
- Hierarchical database structure
  - Zones
  - How is a lookup done
- Caching/prefetching and TTLs
- Reverse name lookup
- What are the steps to creating your own domain?

Next Lecture

- How do forwarding tables get built?
- Routing protocols
  - Distance vector routing
  - Link state routing